

Negative LDO (LT3093) — Working Document

Digital Audio Player · Rev 6

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1. Part Identification

Field	Value
Manufacturer / MPN	Analog Devices — LT3093EDD#TRPBF (package to confirm)
Package	12-lead DFN 3 × 3 mm, exposed pad (pin 13) = IN
Function	Ultralow-noise negative LDO, $-3V9 \rightarrow -3.3$ V analog rail

2. Design Requirements

Requirement	Value
Input	$-3V9$ from inverting charge pump (fed by $+3V9$), -3.9 V nominal
Output	-3.32 V
Load	I/V, summing, OPA1622 (V _− side), DAC analog TBD
Max output	0.6 V RMS (EW300: 119 dB/V worst nozzle, $28\ \Omega \pm 15\%$)
Output current	≈ 27 mA quiescent; ≈ 98 mA pk (200 mA rated)
Current limit	125 mA programmed (R_ILIM)
UVLO	-3.5 V rising
Power good	-3.0 V (90.4% of V_OUT), to MCU GPIO
VIOC	Unused — floated

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1, 2	IN	$-3V9$, C_IN
3	EN/UV	R_EN2 from IN, R_EN1 to GND
4	PG	R_PU to $+3V3$, MCU GPIO
5	PGFB	R_PG1 / R_PG2 midpoint
6	ILIM	R_ILIM to GND (Kelvin to pin 9)
7	VIOC	No connect (float)
8	SET	R_SET C_SET to GND
9	GND	GND
10	OUTS	OUT at load
11, 12	OUT	$-3V3A$, C_OUT, R_PG1
13	Exposed pad	IN ($-3V9$) — own pour and via field, never GND

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
R_SET	33.2 kΩ 0.1%	0402	YAGEO RT0402BRD0733K2L	#1-085
C_SET	2.2 μF (effective TBD)	25 V X7R, low leakage	Murata TBD	—
C_IN	10 μF (6.76 μF eff. at 3.9 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
C_OUT	10 μF (7.56 μF eff. at 3.3 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
R_EN1	50 kΩ 0.1%	0603	YAGEO RT0603BRD0750KL	#1-075
R_EN2	88.7 kΩ 0.1%	0402	YAGEO RT0402BRD0788K7L	—
R_PG1	135 kΩ 0.1% (E192)	0402	YAGEO RT0402BRD07135KL	—
R_PG2	15 kΩ 0.1%	0402	YAGEO RT0402BRD0715KL	#1-078
R_ILIM	15.6 kΩ 0.1% (E192)	0402	YAGEO RT0402BRD0715K6L	—
R_PU	10 kΩ	0603	Generic	—

5. Equations and Results

Output voltage (range over I_{SET} 98 - 102 μA)

$$V_{OUT} = -I_{SET} R_{SET} = -100 \mu A \times 33.2 k\Omega = -3.32 V \quad (-3.254 \text{ to } -3.386 V)$$

Undervoltage lockout (I_{EN} term neglected — ignorable for $R_{EN1} < 100 k\Omega$)

$$V_{IN(UVLO)} = -1.26 \left(1 + \frac{R_{EN2}}{R_{EN1}} \right) - I_{EN} R_{EN2}$$

$$R_{EN2} = R_{EN1} \left(\frac{3.5}{1.26} - 1 \right) = 50 k\Omega \times 1.778 = 88.89 k\Omega \rightarrow 88.7 k\Omega$$

$$V_{IN(UVLO)} = -1.26 \left(1 + \frac{88.7}{50} \right) = -3.495 V \quad (-3.33 \text{ to } -3.69 V)$$

$$V_{IN(UVLO, falling)} = -(1.26 - 0.215) \left(1 + \frac{88.7}{50} \right) = -2.90 V \quad (\text{sim: } \approx -3.0 V)$$

Power good threshold (I_{PGFB} term neglected — ignorable for $R_{PG2} < 30 k\Omega$; R_PG1 is the top resistor)

$$V_{OUT(PG)} = -0.3 \left(1 + \frac{R_{PG1}}{R_{PG2}} \right) - I_{PGFB} R_{PG1}$$

$$R_{PG1} = R_{PG2} \left(\frac{3.0}{0.3} - 1 \right) = 15 \text{ k}\Omega \times 9 = 135 \text{ k}\Omega$$

$$V_{OUT(PG)} = -0.3 \left(1 + \frac{135}{15} \right) = -3.000 \text{ V (90.4\%)} \quad (-2.88 \text{ to } -3.12 \text{ V})$$

Current limit (ILIM regulates to –300 mV; 1.95 k Ω ·A scale factor)

$$I_{LIM} = \frac{1.95 \text{ k}\Omega \cdot A}{R_{ILIM}} = \frac{1.95 \text{ k}\Omega \cdot A}{15.6 \text{ k}\Omega} = 125 \text{ mA} \quad (1.27 \times \text{the } 98 \text{ mA peak})$$

SET noise filter corner (1 μ F effective)

$$f_c = \frac{1}{2\pi R_{SET} C_{SET}} = \frac{1}{2\pi \times 33.2 \text{ k}\Omega \times 1 \mu\text{F}} = 4.8 \text{ Hz}$$

Soft-start (fast start-up to PG threshold –3.0 V, then R_{SET}·C_{SET} tail)

$$\tau = R_{SET} C_{SET}, \quad t_{99} - t_{PG} = \tau \ln \frac{0.32}{0.033} = 2.27 \tau \approx 75 \text{ ms}/\mu\text{F} \quad (2.2 \mu\text{F} : 166 \text{ ms})$$

Maximum SPL (0.6 V RMS, EW300 copper nozzle 119 dB/V)

$$SPL_{MAX} = 119 + 20 \log_{10}(0.6) = 114.6 \text{ dB} \quad (116.6 \text{ dB steel nozzle})$$

Headroom (98 mA peak; datasheet max dropout 280 mV at 100 mA)

$$\Delta V_{PUMP(MAX)} = 3.9 - 3.287 - 0.28 = 0.33 \text{ V} \quad (0.17 \text{ V from the } -3.74 \text{ V step minimum})$$

Dissipation (98 mA pk, $I_{GND} \approx 4 \text{ mA}$)

$$P_D = 0.098 \times (3.981 - 3.32) + 0.004 \times 3.981 \approx 0.08 \text{ W}, \quad \Delta T_J \approx 0.08 \times 34 \approx 2.8^\circ\text{C}$$

Load current (0.6 V RMS = 0.85 V pk design maximum, both channels in phase)

Case	Per channel pk	Signal total	+ Quiescent	LT3093 pk
Full scale, 23.8 Ω (28 Ω –15%)	35.7 mA	71.3 mA	27 mA	$\approx$ 98 mA
Full scale, 28 Ω nominal	30.3 mA	60.6 mA	27 mA	$\approx$ 88 mA
110 dB SPL peak (0.35 V RMS), 23.8 Ω	21.1 mA	42.2 mA	27 mA	$\approx$ 69 mA

Dropout sweep (sim, ideal source; V_OUT 1% low = -3.287 V)

Load	V_IN at dropout	Dropout
29 Ω (≈ 115 mA)	-3.490 V	203 mV
16.5 Ω (≈ 201 mA)	-3.501 V	214 mV

C_SET sweep (sim, ideal -3.9 V source, 29 Ω load; model reads ≈ 0.70 μ V vs datasheet 0.8 μ V at 4.7 μ F, 10 Hz – 100 kHz)

C_SET (effective)	t_99 after PG	Noise, 20 Hz – 20 kHz
0.47 μ F	≈ 35 ms	1.44 μ V RMS
1 μ F	≈ 75 ms	0.75 μ V RMS
2.2 μ F	166 ms (sim)	0.44 μ V RMS
4.7 μ F	≈ 355 ms	0.34 μ V RMS

Output impedance (sim, 29 Ω DC load)

Frequency	Z_OUT
1 kHz	0.26 m Ω
20 kHz	5.2 m Ω
282 kHz (peak)	76 m Ω

Startup and brownout (sim, 2.2 μ F C_SET)

Event	Rising (0 $\rightarrow$ -3.9 V in 20 ms)	Falling (-3.9 $\rightarrow$ 0 V in 50 ms)
Output starts / loses regulation	V_IN = -3.62 V (≈ 0.6 ms after EN trip)	V_IN = -3.490 V
PG changes	V_OUT = -3.000 V, 3.9 ms after start	V_IN = -3.129 V
Settled 99% / shut down	166 ms after PG	V_IN = -2.957 V

Load step (sim, 27 $\leftrightarrow$ 205 mA, 1 μ s edges — conservative vs 98 mA max; identical at -3.9 V and -3.74 V input)

Item	Value
Peak deviation	-7.6 mV / $+3.6$ mV around -3.320 V
Ringing	Damped, ≈ 2 cycles at ≈ 140 kHz
Settling (± 0.3 mV)	≈ 20 μ s
DC shift 27 $\rightarrow$ 205 mA	Not visible (< 0.1 mV)

Line transient and input ripple (sim, 205 mA load)

Item	Value
Line step $-3.9 \rightarrow -3.74$ V (1 μ s edge)	61 μ V peak, settles ≈ 20 μ s (≈ 68 dB)
Line regulation, 160 mV step	≈ 2 μ V DC shift
Input ripple 7.7 mV pp at 1.2 MHz	1.67 μ V pp at OUT (PSRR predicts 0.98 μ V)

6. Important Values

Item	Value
SET pin current	100 μ A (98 – 102 μ A over temp)
EN/UV negative threshold	-1.26 V (-1.20 to -1.33 V), 215 mV hysteresis
PGFB threshold	-300 mV (288 – 312 mV), 7 mV hysteresis
Dropout	214 mV sim / 330 mV max at 200 mA
Peak load	≈ 98 mA at 0.6 V RMS max (49% of rating)
Allowed pump drop at 98 mA	0.33 V from -3.9 V; 0.17 V from -3.74 V (worst case) $\rightarrow R_{OUT} < 1.7 \Omega$
Rated output current	200 mA
Current limit	125 mA programmed; internal backup 400 mA typ, 220 mA min
GND pin current	2.35 mA quiescent; ≈ 7 mA at 200 mA (in dropout)
Shutdown current	3 μ A typ
UVLO divider current	≈ 28 μ A from $-3V9$
PG divider current	≈ 22 μ A from OUT
Output noise	0.44 μ V RMS at 2.2 μ F, 20 Hz – 20 kHz (sim; model $\approx 15\%$ optimistic)
PSRR	-77.9 dB at 1.2 MHz, both -3.9 V and -3.74 V input (sim)
Output impedance	5.2 m Ω at 20 kHz; 76 m Ω peak at 282 kHz (sim)
Load step 27 $\leftrightarrow$ 205 mA	$-7.6 / +3.6$ mV, settles in ≈ 20 μ s; no dropout at -3.74 V (sim)
Line step / ripple	61 μ V for 160 mV step; 1.67 μ V pp from 7.7 mV pp at 1.2 MHz (sim)
Fast start-up	0 $\rightarrow -3.0$ V in 3.9 ms
Firmware delay	Wait ≥ 170 ms after PG before OPA1622 enable (2.2 μ F)
PG at cold start	Reads high until IN passes ≈ -1.8 V — ignore PG until +3V3 is up
Brownout order	Dropout (-3.49 V) $\rightarrow$ PG low (-3.13 V) $\rightarrow$ off (-2.96 V)
C_OUT requirement	≥ 4.7 μ F, ESR < 30 m Ω , ESL < 1.5 nH
θ_{JA} (DFN)	34 $^{\circ}$ C/W
PG in shutdown	Pulled low — reads not-good (opposite of LT3042)
VIOC	Floated; ≈ 120 k Ω internal to GND

#	Problem
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7. Open Problems

#	Problem
1	Charge pump not selected — $R_{OUT} < 1.7 \Omega$ at 98 mA; LM27762 (TI, 2.5Ω typ) meets -3.9 V case only and breaks the LTspice rule; between -3.49 and -3.13 V the rail is out of regulation with PG still good
2	Worst-case UVLO (-3.69 V) leaves ≈ 0.05 V to the -3.74 V step minimum before pump drop
3	Exposed pad is IN ($-3V9$) — dedicated pour and via field; footprint must not tie to GND
4	Quiescent (27 mA) unconfirmed — I/V estimate, OPA1622 I_Q at ± 3.3 V unread
5	DAC analog supply allocation to this rail open
6	C_SET MPN and effective value (SimSurfing); ceramic SET cap is piezoelectric — tap test at Rev A or film
7	Confirm package (DFN vs MSOP-12E) and R_{EN2} / R_{PG1} MPNs
8	PG brownout warning is 2.2 ms at sim ramp rate — real brownout faster; consider hardware mute from PG
9	PG low specified at 100 μ A; 10 k Ω to +3V3 sinks 330 μ A — verify or raise R_{PU}
10	ILIM scale-factor tolerance — confirm minimum limit stays above 98 mA; confirm R_{ILIM} MPN
11	Sims remaining: current limit at 125 mA, PG in shutdown, full chain with pump

8. Revision Log

Rev	Date	Change
0	2026-09-22	Initial issue: R_{SET} , UVLO, PG, soft-start, headroom, dissipation, open problems.
1	2026-09-22	PG divider to 135 k Ω / 15 k Ω , threshold -3.0 V; R_{PG2} reuses #1-078.
2	2026-09-22	Sim results added: dropout, PSRR, noise sweep, $\bar{Z}_{OUT}$, startup, brownout. C_SET 2.2 μ F 25 V. Closed: noise validation, startup knee, UVLO falling behaviour, firmware delay.
3	2026-09-22	Load step result added; closed from remaining sims.
4	2026-09-22	Line transient and 1.2 MHz ripple results added.
5	2026-09-22	Max output set to 0.6 V RMS (EW300); peak load 205 $\rightarrow$ 98 mA. Closed: 200 mA rating exceeded, LTC3440 250 mA overload.
6	2026-09-22	Current limit programmed to 125 mA: R_{ILIM} 15.6 k Ω .